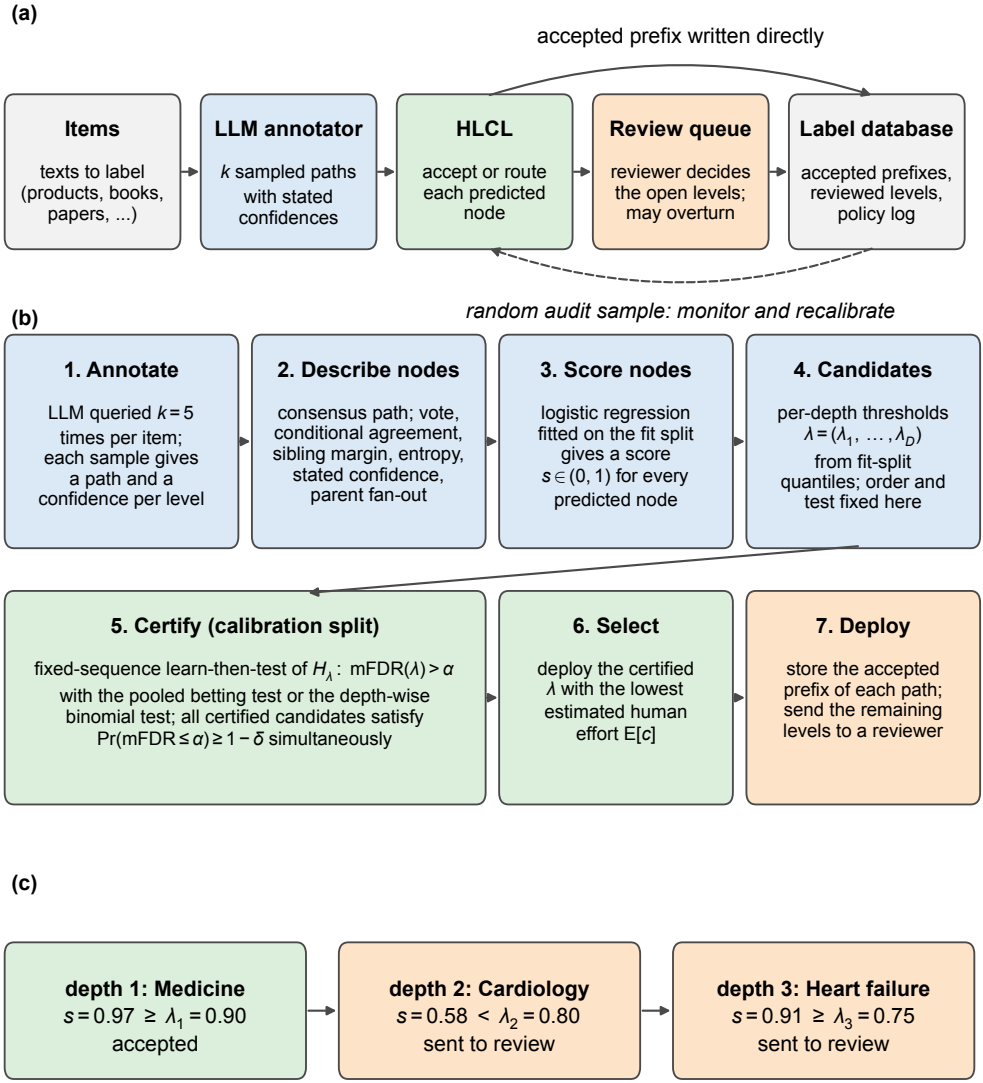

(a)
accepted prefix written directly
Items
texts to label (products, books, papers, ...)
LLM annotator
k sampled paths with stated confidences
HLCL
accept or route each predicted node
Review queue
reviewer decides the open levels; may overturn
Label database
accepted prefixes, reviewed levels, policy log
(b)
random audit sample: monitor and recalibrate
1. Annotate
LLM queried k = 5 times per item; each sample gives a path and a confidence per level
2. Describe nodes
consensus path; vote, conditional agreement, sibling margin, entropy, stated confidence, parent fan-out
3. Score nodes
logistic regression fitted on the fit split gives a score s ∈ (0, 1) for every predicted node
4. Candidates
per-depth thresholds λ = (λ1, … , λD) from fit-split quantiles; order and test fixed here
5. Certify (calibration split)
fixed-sequence learn-then-test of Hλ : mFDR(λ) > α with the pooled betting test or the depth-wise binomial test; all certified candidates satisfy Pr(mFDR ≤ α) ≥ 1 − δ simultaneously
6. Select
deploy the certified λ with the lowest estimated human effort E[c]
7. Deploy
store the accepted prefix of each path; send the remaining levels to a reviewer
(c)
depth 1: Medicine
s = 0.97 ≥ λ1 = 0.90
accepted
depth 2: Cardiology
s = 0.58 < λ2 = 0.80
sent to review
depth 3: Heart failure
s = 0.91 ≥ λ3 = 0.75
sent to review